Roll Number: _____ **Date:** Feb 16, 2026

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

AUXILIARY EXAM

B.E. (III-rd Semester)	**Time**: 05.30 am-8.30 PM
Course Name (Code): Object Oriented Programming (UTA018)	**Duration**: 3 Hours **M. Marks**: 100
Instructor(s) : Raman Kumar Goyal:	

Instructions: Attempt any five questions. Each question is of 20 Marks. Assume missing data, if any, suitably.

Q	Questions	Marks
Q1	(a)Write a C++ program that returns the elements in a vector (STL) that are strictly smaller than their adjacent left and right neighbours. Example: Original Vector elements:1 2 5 0 3 1 7 Vector elements that are smaller than its adjacent neighbours:0 1 (b) Explain three components of STL in detail.	15 05
Q2	Design a C++ program for the following: Write a class template Matrix<T> to represent a matrix of any data type. The class should dynamically allocate memory for a 2D array using pointers in its constructor and properly deallocate memory in the destructor. Include member functions to input and display matrix elements. Overload the * operator to perform matrix multiplication between two matrices using the class template and return the resultant matrix. Also, write a function template multiplyMatrices() that takes two Matrix objects as arguments and performs matrix multiplication (separately from the overloaded operator). In the main() function, create matrices of type int and float, perform multiplication using both the overloaded operator and the function template, and display the results.	20
Q3	(a)What is a friend function? Write a C++ program to demonstrate a friend function. (b) Explain encapsulation and polymorphism with the help of examples in brief.	12 08
Q4	(a)Explain different types of inheritances in detail. (b)Explain the concept of static member function in brief.	15 5
Q5	What are constructors? Explain different types of constructors used in C++ with examples.	20
Q6	(a)Write a C++ program to demonstrate run time polymorphism using a base class and a derived class where a virtual function is overridden in the derived class and called using a base class pointer. Show how dynamic binding occurs at runtime by invoking the function through the base class reference or pointer. (b) Explain exception handling in C++ with the help of an example.	12 8

Viewing of answer sheets is scheduled on 19 February 2:20PM in L-506 . No request of change in timings will be considered.